

Final Exam

Due: 3 pm June 15, 2022

Name _____

Student ID _____

VERSION E

Instructions:

- NO interactions and communications with anyone in any form—**academic integrity and honesty** is important
- Answer all questions using a blue/black pen on A4 papers
- Upload your answers and the first page of the exam to WM5 by **3:20 pm** (20 points off if uploading after **3:20 pm**, and any submission after **3:30 pm** will not be accepted)
- Read and sign the statement below

By signing this, I declare that the submitted exam was produced independently by me, without help from others.

Signature

Microeconomics **Final Exam**, Spring 2022

Name and Student ID:

Show all your work. You will have the entire class period to complete this exam. This exam is "closed book" and "closed note". Good luck.

1. [20 points] A crime has been committed and two suspects have been arrested. The suspects are taken to the police station and the district attorney meets with each one separately. To each she makes the following offer: "If you each confess, I'll send you both to jail for 5 years. If neither of you confesses, I can still get you on a lesser charge and send you to jail for 2 years each. If your buddy confesses and you don't, you'll get 10 years and he'll get 1. But if *you* are the only one to confess, you'll get off with 1 year while I put *him* away for 10.

- a. [9 points] Write down the normal-form game that specifies the players, possible strategies, and payoffs.

B

	Confess	not confess
A Confess	$(-5, -5)$	$(-1, -10)$
not confess	$(-10, -1)$	$(-2, -2)$

- b. [6 points] Does either of the suspects have a dominant strategy? What is the Nash Equilibrium of the game?

Their dominant strategy are confess, because the payoff of confess is larger regardless of the other player's strategy

$$N.E. (confess, confess) \Rightarrow (-5, -5)$$

- c. [5 points] If the game is played infinitely, and both players play grim strategy, at what values of δ would the two suspects cooperate?

$$\begin{aligned}
 E_A(\text{confess in } t \text{ period}) &= -1 + (-5)\delta + (-5)\delta^2 + \dots \\
 &= -1 + (-5)\delta(1 + \delta + \delta^2 + \dots) \\
 &= -1 + (-5)\delta \cdot \frac{1}{1-\delta} \\
 E_A(\text{not confess from } t \text{ to infinity}) &= -2 + (-2) \cdot \delta + (-2) \cdot \delta^2 + \dots = (-2)(1 + \delta + \delta^2 + \dots) = (-2) \cdot \frac{1}{1-\delta} \\
 \text{if cooperate} &\Rightarrow (-2) \cdot \frac{1}{1-\delta} > -1 + (-5) \cdot \frac{1}{1-\delta} \Rightarrow \delta > \frac{1}{4}
 \end{aligned}$$

Handwritten notes on the left:

- $\frac{-2}{1-\delta} + 1 + \frac{5\delta}{1-\delta} > 0$
- $\Rightarrow \frac{-2+1-\delta+5\delta}{1-\delta} > 0$
- $\Rightarrow \frac{-1+4\delta}{1-\delta} > 0$

2. [10 points] Lucy and Ricky are making plans for Saturday night. They can go to either a ballet or a boxing match. Each will make the choice independently, although as you can see from the following table, there are some benefits if they end up doing the same thing.

		Ricky	
		Ballet	Boxing Match
Lucy	Ballet	100, 30	-90, -90
	Boxing Match	-90, -90	30, 100

Find the Nash equilibria of the game (including a mixed strategy Nash equilibrium).

pure strategy: (Ballet, Ballet), (Boxing, Boxing)

mixed strategy:

$$E(\pi_L | B_R) = 100q + (-90)(1-q) = 100q - 90 + 90q = 190q - 90$$

$$E(\pi_L | B_o) = -90q + (-90)(1-q) = -90q - 90 + 90q = -90$$

BR_L:

$$\begin{aligned}
 190q - 90 &> -90 \Rightarrow 190q > 0 \Rightarrow q > 0 \\
 190q - 90 &< -90 \Rightarrow 190q < 0 \Rightarrow q < 0 \\
 190q - 90 &= -90 \Rightarrow 190q = 0 \Rightarrow q = 0
 \end{aligned}$$

$$E(\pi_R | B_R) = 30p + (-90)(1-p) = 120p - 90$$

$$E(\pi_R | B_o) = -90p + 100(1-p) = 100 - 190p$$

BR_R:

$$\begin{cases} 120p - 90 > 100 - 190p \Rightarrow 310p > 190 \Rightarrow p > \frac{19}{31} \Rightarrow q = 1 \\ p < \frac{19}{31} \Rightarrow q = 0 \\ p = \frac{19}{31} \Rightarrow q = 0 \sim 1 \end{cases}$$

$\Rightarrow$ mixed strategy $\Rightarrow \left(\frac{19}{31} \text{ Bullet} + \frac{12}{31} \text{ Boxing match}, \frac{12}{31} \text{ Bullet}, \frac{19}{31} \text{ Boxing match} \right)$

3. [30 points] Suppose that there are two producers of prokrypton-B, a rare mineral capable of incapacitating aliens from distant planets. Governments across the globe are interested in purchasing defensive stores of the rare mineral. The market demand for prokrypton-B is given by $P = 200 - 0.2Q$, where $Q = q_A + q_B$ is the collective output of the industry, in tons; q_A is the output of Firm A; and q_B is the output of Firm B. Assume that each firm produces prokrypton-B at a marginal cost of \$10 per ton, and compete by quantity (Cournot).

a. [6 points] Derive each firm's reaction function.

b. [6 points] Solve for the equilibrium quantity of each firm, as well as industry output (Q).

- c. [6 points] Solve for the market price of prokrypton-B, as well as the profit earned by each firm.
- d. [12 points] Suppose that a labor dispute drives Firm B's cost to \$12 per ton.
- i. What happens to the output of each firm as a result? (Solve for the new output)
 - ii. What is the new market price of prokrypton-B, and what is the new profit for each firm?

4. [20 points] Consider a two-firm industry producing two differentiated products indexed by $i = 1, 2$. For simplicity, we assume that production is costless. Suppose that the demand functions for the two products are given by $q_1 = 168 - 2p_1 + p_2$ and $q_2 = 168 - 2p_2 + p_1$.
- a. [10 points] Suppose that the two firms set their price independently and simultaneously. Find the Bertrand-Nash equilibrium.
- b. [10 points] Suppose that the two firms move sequentially rather than simultaneously. In particular, suppose that firm 1 sets its price first. After observing firm 1's selection, firm 2 chooses its price. Find the Subgame Perfect Nash Equilibrium.

5. [24 points] Suppose that two firms are competing for a government contract, and each firm has a 50 percent chance of winning. The stock of the firm that wins the contract will be worth \$40, while the stock of the loser will be worth \$10. Calculate the expected value and variance of the two shares for the following three cases.
- a. [8 points] If only one firm can win, what are the expected value and variance of buying one share of each firm.
- b. [8 points] If the government awards contracts only to both firms or to neither firm, what are the expected value and variance of buying one share of each firm?

- c. [8 points] Redo part a and b for the expected value and variance of buying two shares of one firm.
6. [16 points, 8 each] Answer the following two questions on risk and uncertainty.
- a. You have a utility function given by $u(I) = 10 \ln I$, where I represents the monetary payoff from an investment. You are considering making an investment which, if it pays off, will give you a payoff of \$100,000, but if it fails, it will give you a payoff of \$20,000. Each outcome is equally likely. What is the risk premium for this lottery?

- b. Consider a market of risk-averse decision makers, each with a utility function $u(I) = \sqrt{I}$, where I is income. Each decision maker has an income of \$90,000, but faces the possibility of a catastrophic loss of \$50,000 in income. Each decision maker can purchase an insurance policy that fully compensates her for her loss. This insurance policy has a cost of \$5,900. Suppose each decision maker potentially has a different probability q of experiencing the loss. What is the smallest value of q so that a decision maker purchases insurance?